Quantifying and Predicting Flight Delay Propagation

Using Statistical Learning Methods

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Presentation Roadmap

- **Problem and Motivation**
  Why flight delay propagation matters
- **Data and Feature Engineering**
  Dataset, cleaning, and derived features
- **Exploratory Data Analysis**
  Key patterns and visual insights
- **Statistical Modeling**
  Classification & regression approaches
- **Results and Conclusions**
  Findings, limitations, and future work

The Problem

Research Question

To what extent do delays propagate between sequential flights of the same aircraft, and which factors best predict downstream delay?

When an aircraft arrives late, the next flight it operates is often delayed too — a cascading effect.

Airlines, airports, and passengers bear the cost. Understanding which factors drive propagation could help mitigate it.

We use statistical learning methods from STAT 654 to quantify and predict this phenomenon.

Data Overview

5.8M	14	322	~500K
Total Flights	**Airlines**	**Airports**	**Modeled Flights**
U.S. domestic, 2015	Major U.S. carriers	Origin airports	After cleaning & linking

Source: Kaggle — 2015 U.S. Flight Delays and Cancellations

Key Steps:

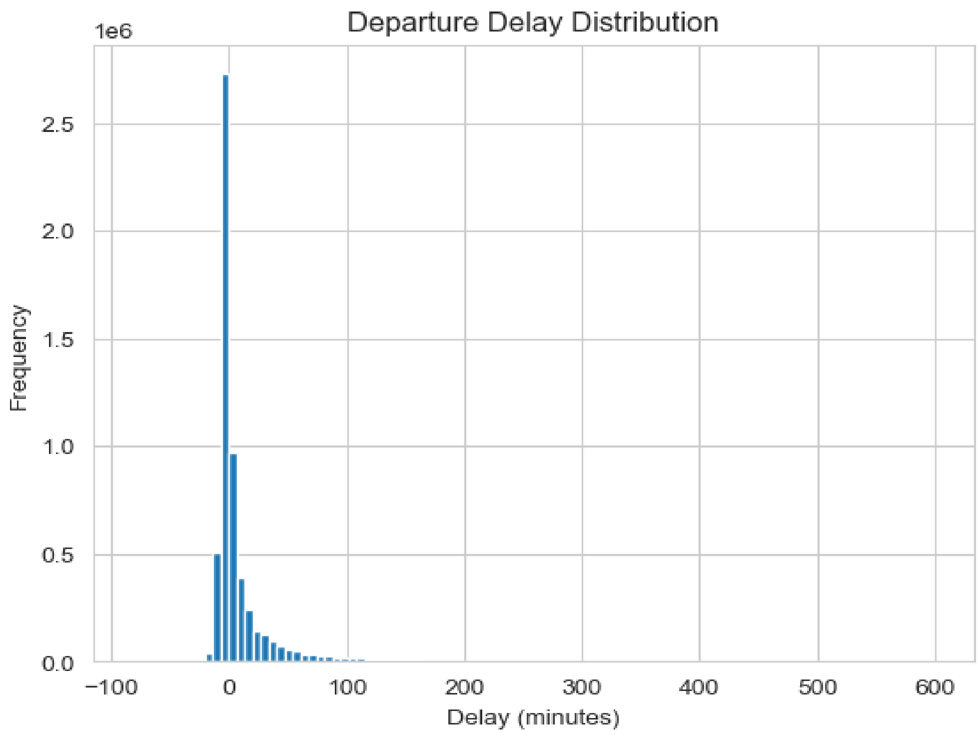
- Tracked aircraft by tail number to link sequential flights
- Converted HHMM times to datetime for accurate turnaround calculations
- Filtered invalid sequences (negative turnaround, mismatched airports)
- Capped extreme delays at 600 minutes to reduce distortion

Feature Engineering

Feature	Description	Category
`prev_arrival_delay`	Minutes late the previous flight arrived	**Propagation**
`turnaround_time`	Time between arrival and next departure (minutes)	**Propagation**
`airport_congestion`	Average delay at that airport/hour combination	**Environment**
`hour`	Hour of scheduled departure (0–23)	**Temporal**
`DAY_OF_WEEK`	Day of the week (1=Mon – 7=Sun)	**Temporal**
`AIRLINE`	Carrier code (14 airlines, one-hot encoded)	**Operational**

Response variable: delay_indicator (1 if departure delay > 15 min, else 0) and DEPARTURE_DELAY (minutes)

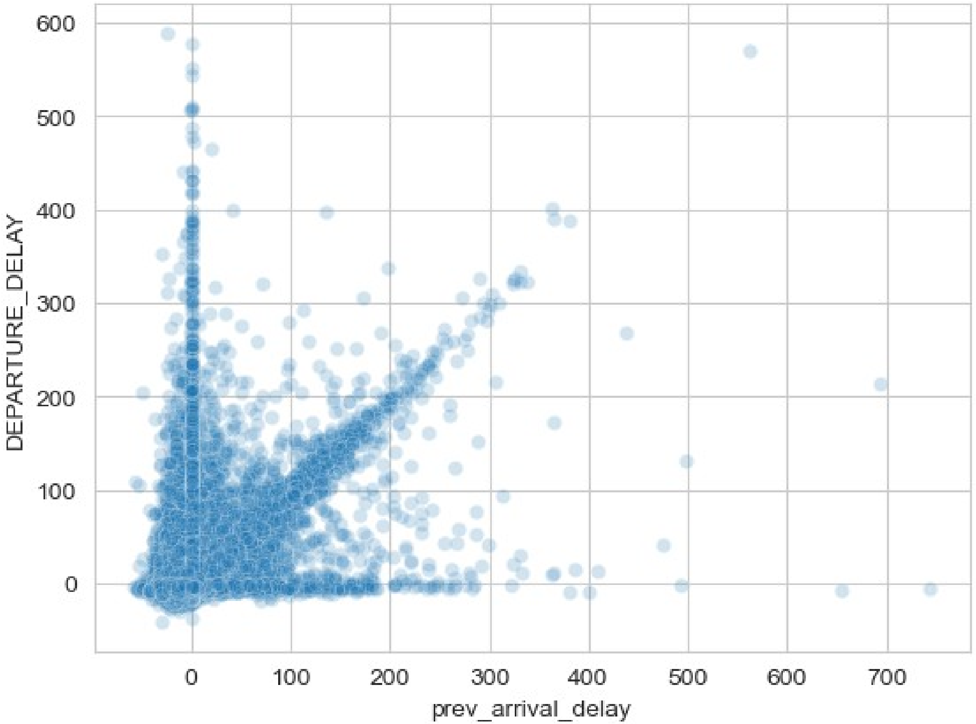
Departure Delay Distribution



Key Takeaways

- Heavily right-skewed: most flights depart on time or early
- 82.3% of flights are not delayed (delay ≤ 15 min)
- Only 17.7% classified as delayed
- Class imbalance must be considered in modeling

Delay Propagation Is Real



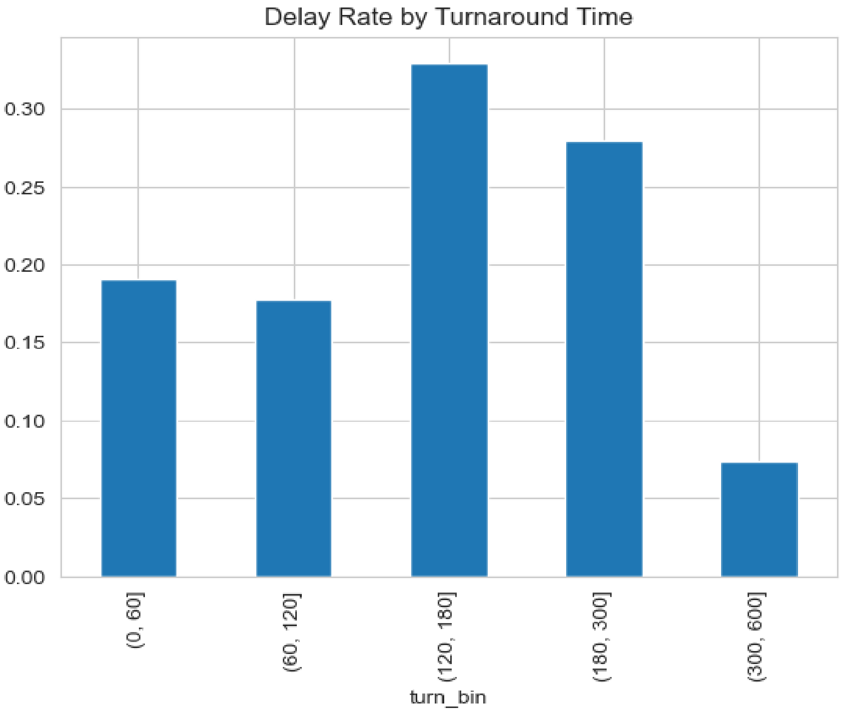
70% chance of outbound delay when inbound is delayed (>15 min)

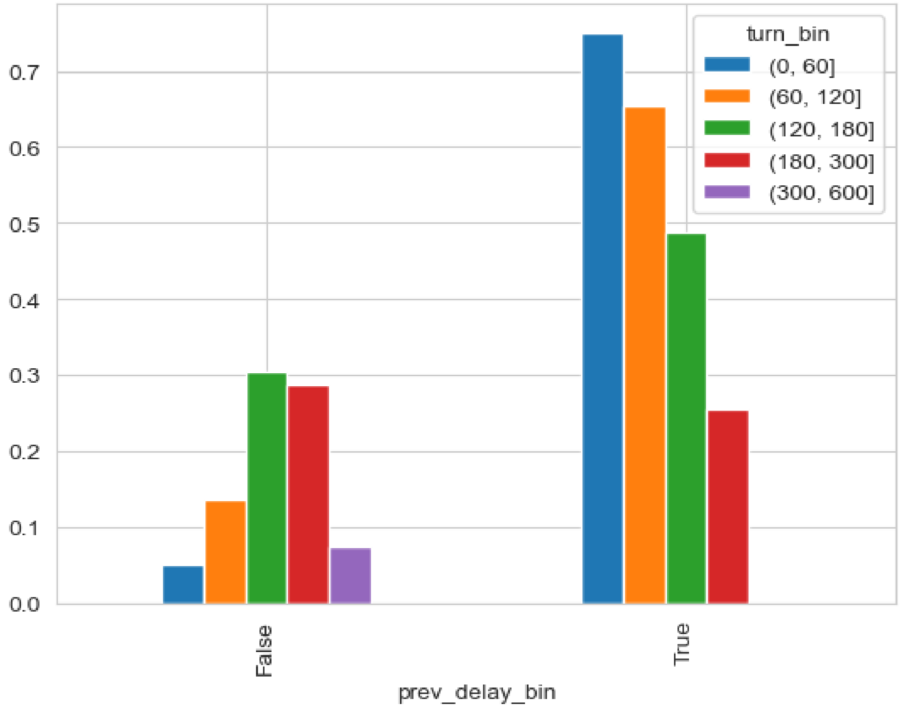
vs. only 10% when inbound is on time

r = 0.469 Correlation between inbound & outbound delay

Moderate correlation suggests a nonlinear relationship — other factors at play

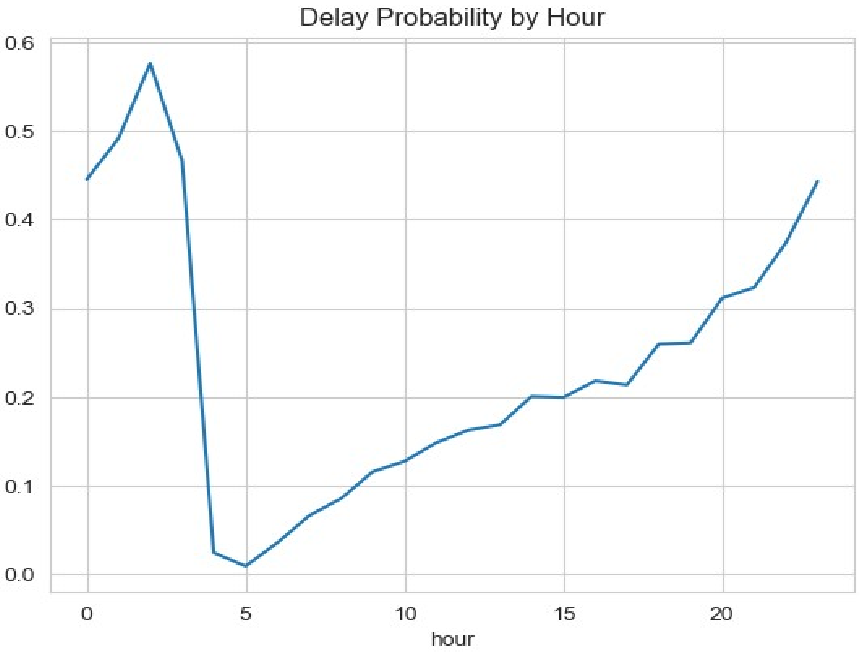
Turnaround Time and Delay Interaction

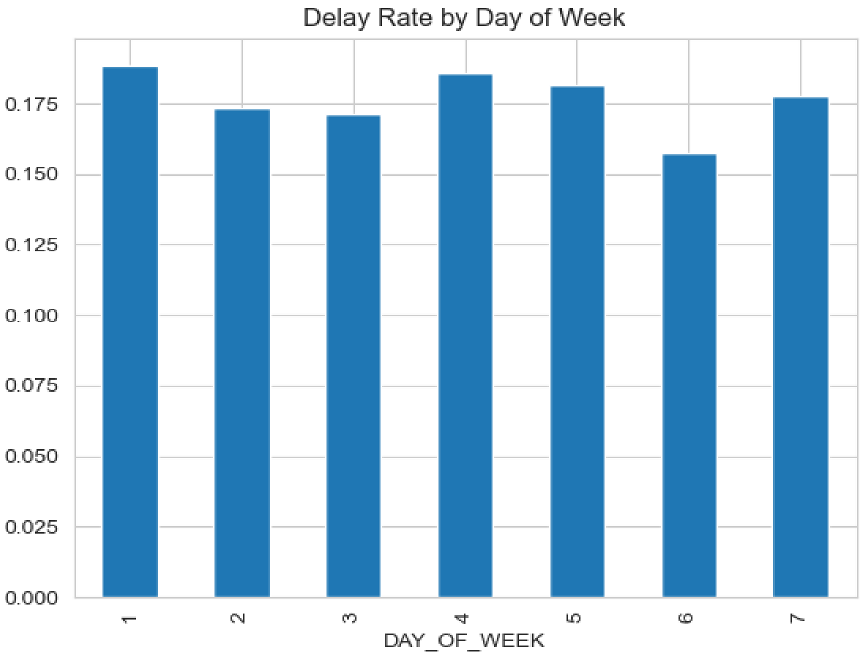




Moderate turnaround windows (120–300 min) show the highest delay rate. Very long turnarounds (300–600 min) allow recovery. When the inbound flight is delayed, short turnarounds amplify risk the most.

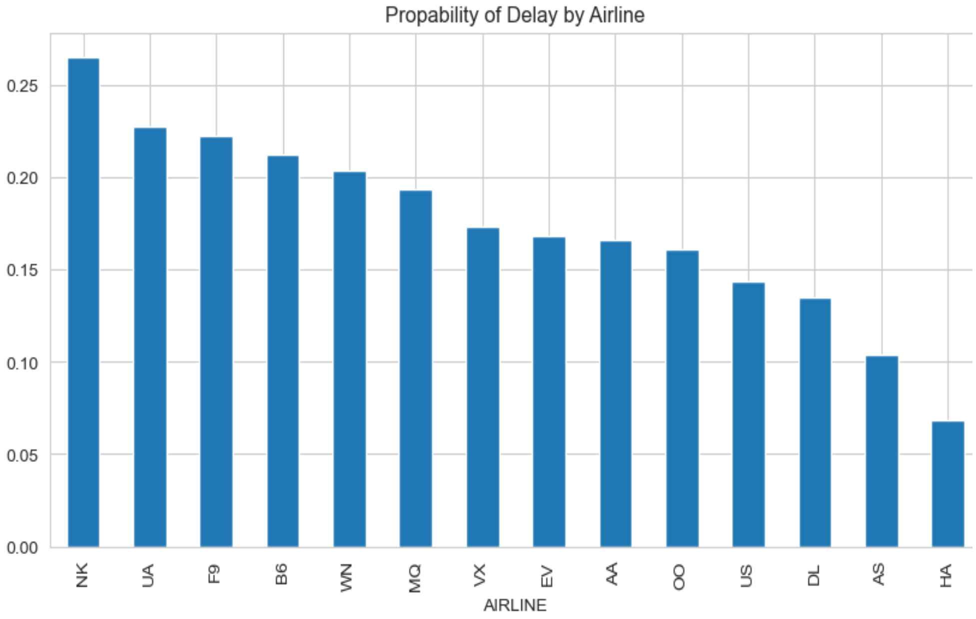
Temporal Patterns in Delays

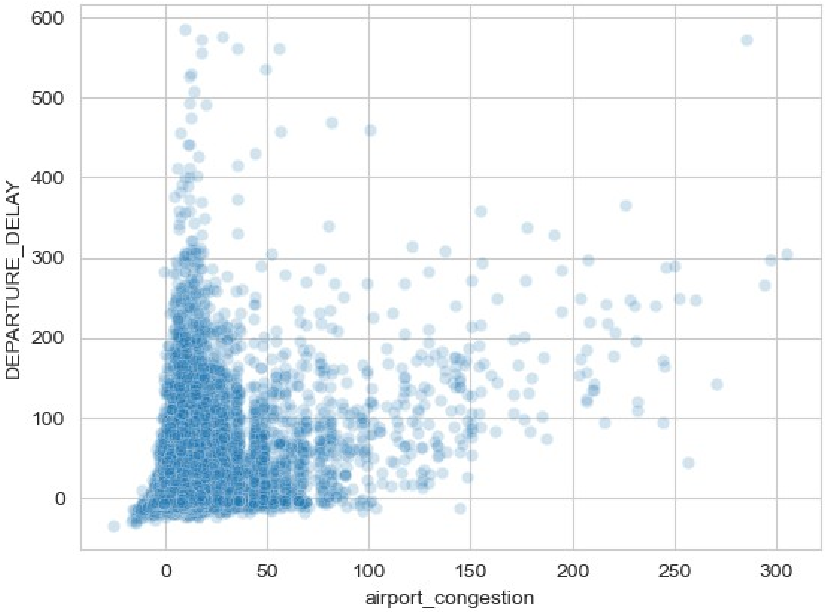




Delays build throughout the day, peaking in late evening. Early-morning flights have the lowest delay risk. Day-of-week differences are minor (~15–18%), with Saturday slightly lower.

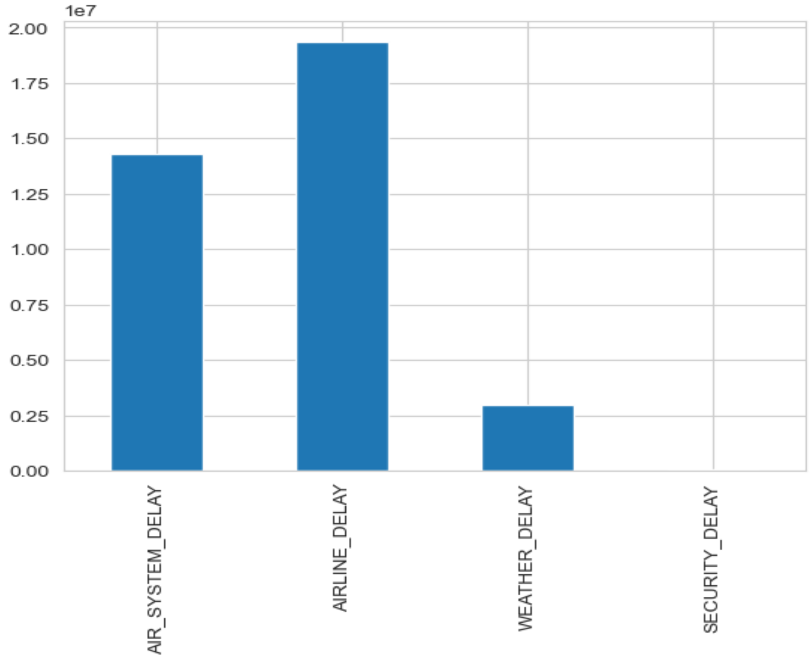
Airline and Airport Congestion Effects

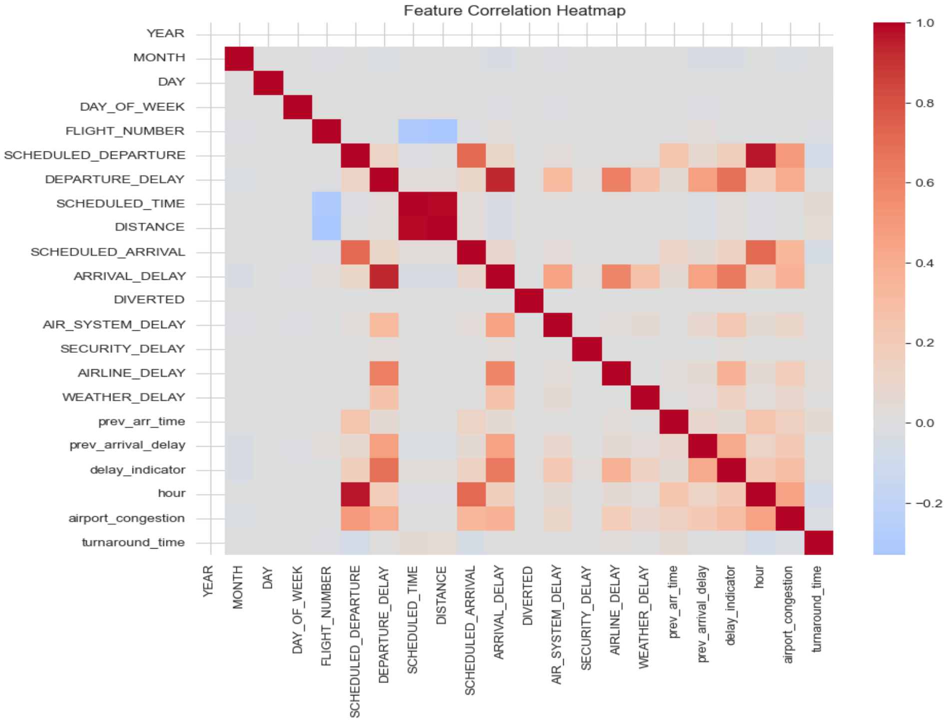




Delay probability varies widely by airline (7% for HA to 27% for NK). Airport congestion shows a positive relationship with departure delay, motivating its inclusion as a feature.

Delay Causes and Feature Correlations





Airline-related and air-system delays dominate; weather and security are minimal. The heatmap confirms prev_arrival_delay and airport_congestion as strong correlates of departure delay.

Modeling Strategy

Classification

Will this flight be delayed (>15 min)?

- Baseline: prev_arrival_delay only
- Baseline+1: + turnaround_time
- Full Model: all 6 predictors + airline dummies
- Lasso Logistic: L1 regularization for variable selection

Regression

How many minutes will it be delayed?

- OLS Linear Regression with all predictors
- Lasso Linear: L1 regularization to check for overfitting

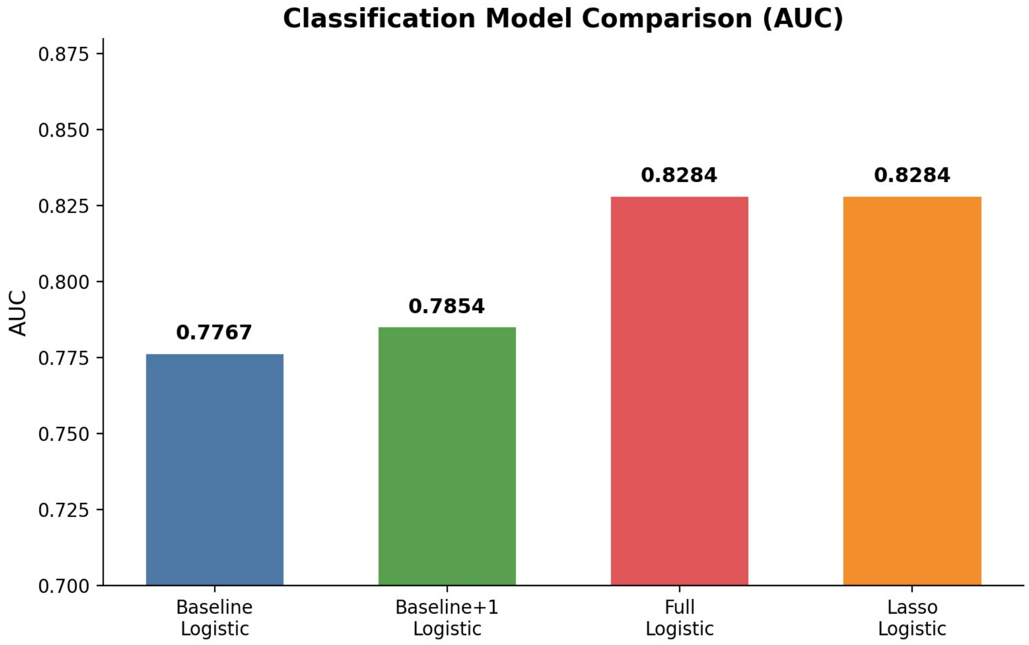
Metrics: RMSE, R^2

80/20 train-test split throughout

Classification: Building Up the Model

Baseline	Baseline + 1	Full Model
0.777	**0.785**	**0.828**
AUC	AUC	AUC
86.3% accuracy	86.3% accuracy	86.9% accuracy
`log(p/(1-p)) = -1.78 + 0.048 × prev_delay`	`+ turnaround_time`	`+ congestion, hour, day, airline`
Catches only 28% of delays	*Slight AUC improvement*	*Catches 35% of delays*

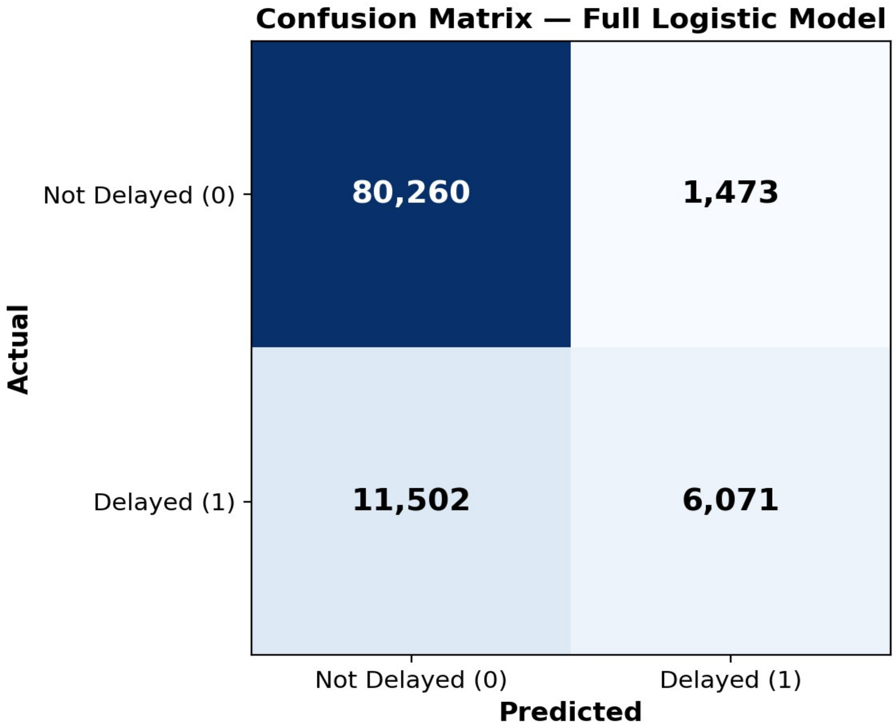
Classification Model Comparison



Interpretation

- AUC jumped from 0.777 → 0.828 as more predictors were added
- Lasso confirmed no overfitting (identical AUC to full model)
- Biggest gain came from adding congestion, hour, and airline
- All predictors significant at $p < 0.05$

Confusion Matrix — Full Logistic Model



Key Metrics

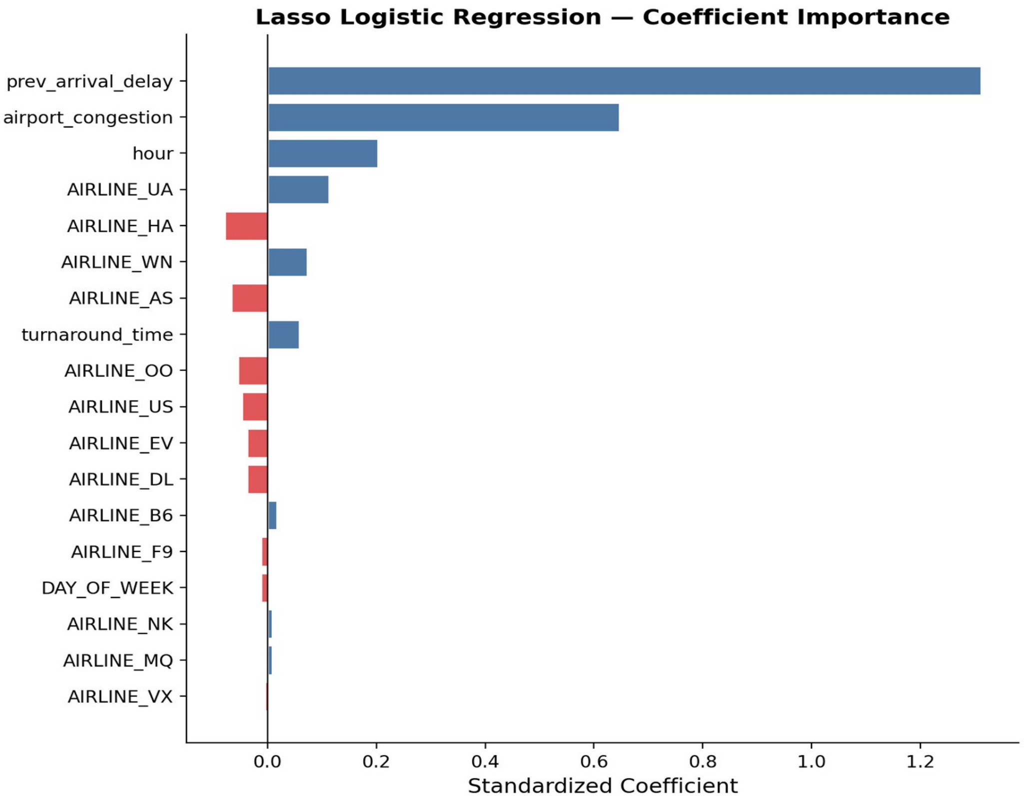
Accuracy: 86.9%

Sensitivity (recall): 34.5% — catches about 1 in 3 actual delays

Specificity: 98.2% — excellent at identifying on-time flights

The model is conservative: it rarely falsely predicts a delay, but misses many actual delays due to class imbalance (82% vs 18%)

Which Predictors Matter Most?



Lasso Results

- prev_arrival_delay is the dominant predictor by far
- airport_congestion and hour are the next most important
- No coefficients zeroed out — all features contribute
- DAY_OF_WEEK near zero (-0.011), least important
- United Airlines (UA) most delay-prone; Hawaiian (HA) least

Regression: Predicting Delay Minutes

OLS Coefficients

Predictor	Coefficient	p-value
prev_arrival_delay	0.466	<0.001
airport_congestion	0.779	<0.001
AIRLINE_UA	4.385	<0.001
hour	-0.030	0.004
turnaround_time	0.001	<0.001

$R^2 = 0.319$

- Linear regression explains ~32% of variation in delay minutes
- Remaining 68% of variation reflects unmeasured factors (weather details, mechanical issues, crew scheduling)

RMSE = 28.1 min

- Predictions are off by about half an hour on average

- Lasso zeroed out airlines B6, MQ, and VX
  - no additional explanatory power

Model Performance Summary

Model	Type	AUC	Accuracy	RMSE	R^2
Baseline Logistic	Classification	0.7767	86.29%	—	—
Baseline+1 Logistic	Classification	0.7854	86.30%	—	—
Full Logistic	Classification	0.8284	86.91%	—	—
Lasso Logistic	Classification	0.8284	86.91%	—	—
Linear Regression	Regression	—	—	28.15	0.319
Lasso Linear	Regression	—	—	28.15	0.317

Conclusions

- **Propagation is real and measurable**

  Flights following a delayed inbound aircraft have a 7× higher chance of being delayed themselves.

- **prev_arrival_delay is the strongest predictor**

  Both classification and regression models show it dominates all other features by a wide margin.

- **Full model achieves AUC of 0.830**

  Adding congestion, time-of-day, and airline significantly improved predictions over the baseline.

- **Challenges remain**

  32% R^2 in regression means most delay variation is still unexplained — better data (weather, crew, mechanical) could help.

Future Work

- Incorporate granular weather data (METAR/ASOS) as additional features
- Explore nonlinear methods (Random Forest, Gradient Boosting) to improve R^2
- Address class imbalance with SMOTE or adjusted thresholds
- Extend to multi-hop propagation (3+ sequential flights)
- Test on more recent data (post-COVID) to assess generalizability

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Thank you!

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